

3 Raffinose is a sugar found in many plant species.

(a) Figure 3.1 shows the structure of one molecule of raffinose.

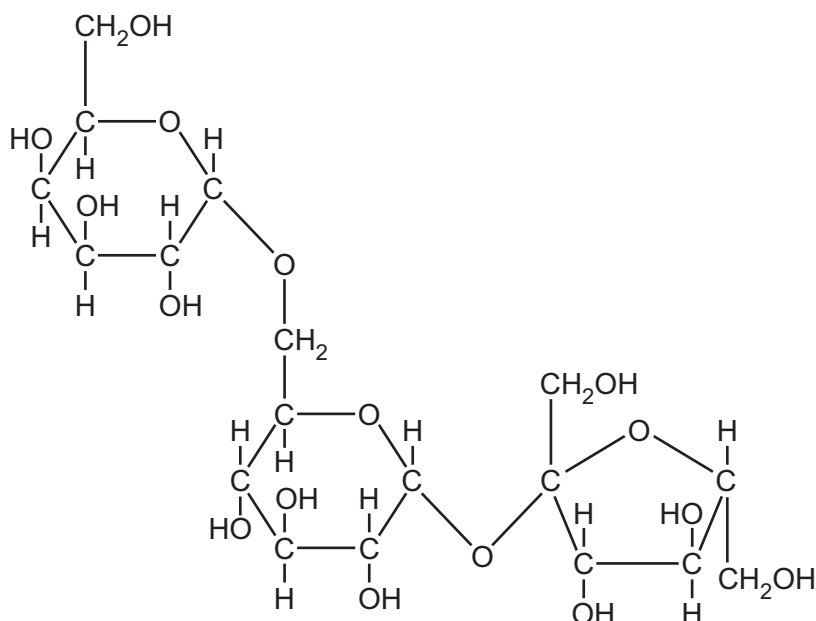


Figure 3.1

- (i) Raffinose is formed by the reaction of three different monosaccharides, one of which is glucose.

On Figure 3.1, draw around the part of the molecule that forms glucose when the glycosidic bonds in raffinose are hydrolysed.

[1]

- (ii) State the type of glucose molecule that is formed when the glycosidic bonds of raffinose are hydrolysed.

..... [1]

- (iii) Raffinose produces the same results as sucrose when it is tested with Benedict's solution.

Complete Table 3.1 to show some features of the sugars glucose, maltose and raffinose. Maltose is a disaccharide.

Table 3.1

sugar	number of glycosidic bonds in one molecule	reducing sugar or non-reducing sugar
glucose		
maltose		
raffinose		

[3]

- (b) Some bacteria in the human digestive system synthesise an enzyme known as α -GAL, which breaks down raffinose.

The rate of breakdown of raffinose by α -GAL at different concentrations of raffinose is shown in Figure 3.2.

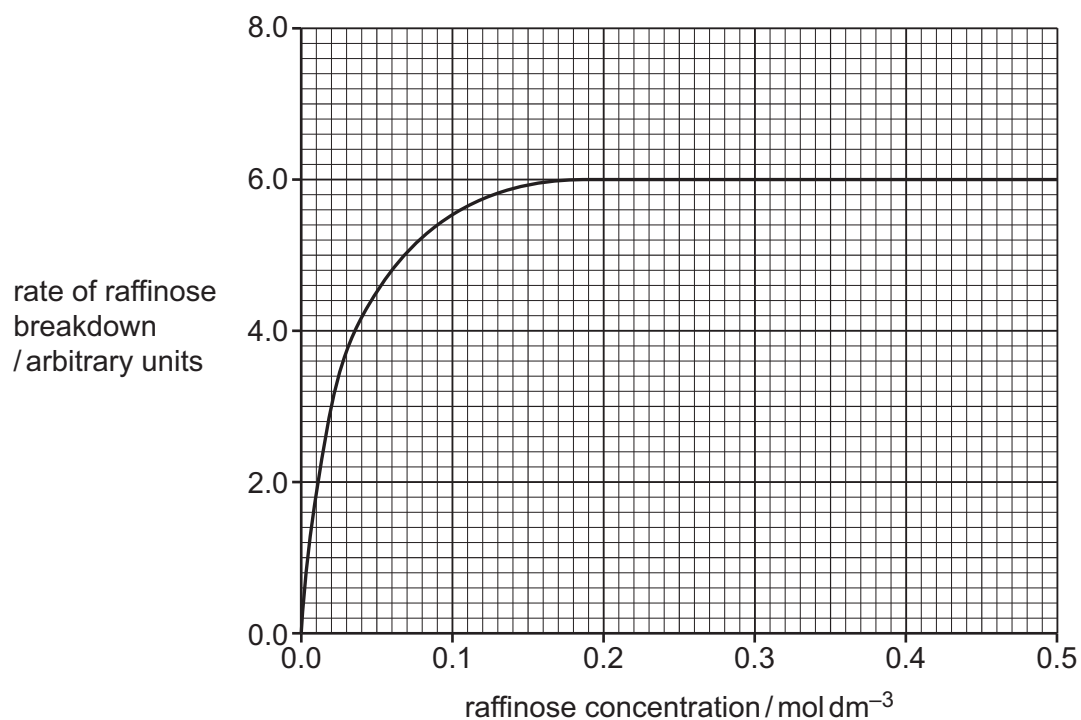


Figure 3.2

Determine the Michaelis–Menten constant, K_m , for α -GAL using the data in Figure 3.2.

$K_m = \dots\dots\dots$ [2]

- (c) One of the molecules formed from the breakdown of raffinose by bacteria is shown in Figure 3.3.

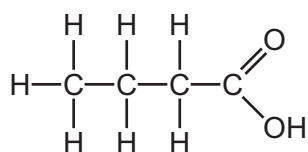


Figure 3.3

- (i) State the type of molecule shown in Figure 3.3.

$\dots\dots\dots$ [1]



(ii) The molecule shown in Figure 3.3 can be used to make a triglyceride.

Describe how this molecule can become part of a triglyceride molecule.

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..... [3]

[Total: 11]